MONETARY POLICY TOOLS AND MONETARY TRANSMISSION

Learning Outcome

- describe tools used to implement monetary policy tools and the monetary transmission mechanism, and explain the relationships between monetary policy and economic growth, inflation, interest, and exchange rates

Central banks have three primary tools available to them: open market operations, the refinancing rate, and reserve requirements.

Open Market Operations

One of the most direct ways for a central bank to increase or reduce the amount of money in circulation is through **open market operations**. Open market operations involve the purchase and sale of government bonds from and to commercial banks or designated market makers. For example, when the central bank buys government bonds from commercial banks, this increases the reserves of private sector banks on the asset side of their balance sheets. If banks then use these surplus reserves by increasing lending to corporations and households, then broad money growth expands through the money multiplier process. Similarly, the central bank can sell government bonds to commercial banks. In so doing, the reserves of commercial banks decline, reducing their capacity to make loans (i.e., create credit) to households and corporations and thus causing broad money growth to decline through the money multiplier mechanism. In using open market operations, the central bank may target a desired level of commercial bank reserves or a desired interest rate for these reserves.

The Central Bank's Policy Rate

The most obvious expression of a central bank's intentions and views comes through the interest rate it sets. The name of the **official interest rate** (or **official policy rate** or just **policy rate**) varies from central bank to central bank, but its purpose is to influence short- and long-term interest rates and ultimately real economic activity.

The interest rate that a central bank sets and that it announces publicly is normally the rate at which it is willing to lend money to the commercial banks (although practices do vary from country to country). This policy rate can be achieved by using short-term collateralized lending rates, known as repo rates. For example, if the central bank wishes to increase the supply of money, it might buy bonds (usually government bonds) from the banks, with an agreement to sell them back at some time in the future. This transaction is known as a **repurchase agreement**. Normally, the maturity of repo agreements ranges from overnight to two weeks. In effect, this represents a secured loan to the banks, and the lender (in this case the central bank) earns the repo rate.

Suppose that a central bank announces an increase in its official interest rate. Commercial banks normally would increase their **base rates** at the same time. A commercial bank's base rate is the reference rate on which it bases lending rates to all other customers. For example, large corporate clients might pay the base rate plus 1 percent on their borrowing from a bank, whereas the same bank might lend money to a small corporate client at the base rate plus 3 percent. But why would commercial banks immediately increase their base or reference rates just because the central bank's refinancing rate had increased?

The answer is that commercial banks do not want to lend at a rate of interest below that which they are charged by the central bank. Effectively, the central bank can force commercial banks to borrow from it at this rate because it can conduct open market operations that create a shortage of money, forcing the banks to sell bonds to it with an agreed-upon repurchase price (i.e., a repurchase agreement). The repo rate would be such that the central bank earned the official refinancing rate on the transactions.

The name of each central bank's official refinancing rate varies. The Bank of England's refinancing rate is the **two-week repo rate**. In other words, the Bank of England fixes the rate at which it is willing to lend two-week money to the banking sector. The European Central Bank's (ECB's) official policy rate is known as the **refinancing rate**, which defines the rate at which the ECB is willing to lend short-term money to the Euro area banking sector.

The corresponding rate in the United States is the discount rate, which is the rate for member banks borrowing directly from the Federal Reserve System. The most important interest rate used in US monetary policy is the **federal funds rate**. The federal funds rate (or **fed funds rate**) is the interbank lending rate on overnight borrowings of reserves. The Federal Open Market Committee (FOMC) seeks to move this rate to a target level by adjusting administered rates primarily the interest on reserve balances and overnight reverse repurchase agreement facility rate. The level of the rate is reviewed by the FOMC at its meetings held every six weeks (although the target can be changed between meetings, if necessary).

Through the setting of a policy rate, a central bank can manipulate the amount of money in the money markets. Generally speaking, the higher the policy rate, the higher the potential penalty that banks will have

The third primary way in which central banks can limit or increase the supply of money in an economy is through the **reserve requirement**. We already have seen that the money creation process is more powerful the lower the percentage reserve requirement of banks. So, a central bank could restrict money creation by raising the reserve requirements of banks. However, this policy tool is not used much today in developed market economies. Indeed, some central banks, such as the Bank of England, no longer even set minimum reserve requirements for the banks under their jurisdiction. Changing reserve requirements frequently is disruptive for banks. For example, if a central bank increased the reserve requirements, a bank that was short on reserves might have to cease its lending activities until it had built up the necessary reserves, because deposits would be unlikely to rise quickly enough for the bank to build its reserves in this way. However, reserve requirements are still actively used in many emerging market countries to control lending and remain a potential policy tool for those central banks that do not currently use it.

To summarize, central banks can manipulate the money supply in one of three ways:

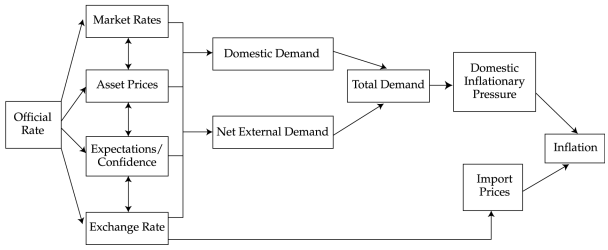
- open market operations;
- official policy rates and associated actions in the repo market; and
- manipulation of official reserve requirements.

The Transmission Mechanism

The overarching goal of a central bank is to maintain price stability. We have demonstrated how a central bank can manipulate the money supply and growth of the money supply. We also indicated how policy rates set and targeted by the central banks are usually very short term in nature; often they target overnight interest rates. However, most businesses and individuals in the real economy borrow and lend over much longer time frames than this. It may not be obvious, then, how changing short-term interest rates can influence the real economy, particularly if money neutrality holds in the long run. The fact that central bankers believe that they can affect real economic variables, in particular economic growth, by influencing broad money growth suggests that they believe that money is not neutral—at least not in the short run.

Exhibit 3 presents a stylized representation of the **monetary transmission mechanism**. This is the process whereby a central bank's interest rate is transmitted through the economy and ultimately affects the rate of increase of prices—that is, inflation.

Exhibit 3: A Stylized Representation of the Monetary Transmission Mechanism



Source: Bank of England, https://www.bankofengland.co.uk/.

Suppose that a central bank announces an increase in its official interest rate. The implementation of the policy may begin to work through the economy via four interrelated channels. Those channels include bank lending rates, asset prices, agents' expectations, and exchange rates. First, as described earlier, the base rates of commercial banks and interbank rates should rise in response to the increase in the official rate. Banks, in turn, would increase the cost of borrowing for individuals and companies over both short- and long-term horizons. Businesses and consumers would then tend to borrow less as interest rates rise. An increase in short-term interest rates could also cause the price of such assets as bonds or the value of capital projects to fall as the discount rate for future cash flows rises.

Market participants would then come to the view that higher interest rates will lead to slower economic growth, reduced profits, and reduced borrowing to finance asset purchases. Exporters' profits might decline if the rise in interest rates causes the country's exchange rate to appreciate, because this would make domestic exports more expensive to overseas buyers and dampen demand to purchase them. The fall in asset prices as well as an increase in prices would reduce household financial wealth and therefore lead to a reduction in consumption growth. Expectations regarding interest rates can play a significant role in the economy. Often companies and individuals will make investment and purchasing decisions based on their interest rate expectations, extrapolated from recent events. If the central bank's interest rate move is widely expected to be followed by other interest rate increases, investors and companies will act accordingly. Consumption, borrowing, and asset prices may all decline as a result of the revision in expectations.

A rise in the central bank's policy rate can reduce real domestic demand and net external demand (that is, the difference between export and import consumption) through a wide range of interconnected ways. Weaker total demand would tend to put downward pressure on the rate of domestic inflation—as would a

interconnected channels:

- Short-term interest rates;
- Changes in the values of key asset prices;
- The exchange rate; and
- The expectations of economic agents.

Question Set

1. Which of the following variables are *most likely* to be affected by a change in a central bank's policy rate?

A. Asset prices only
B. Expectations about future interest rates only
C. Both asset prices and expectations about future interest rates

Hide Solution

Solution:

C is correct. The price of equities, for example, might be affected by the expectation of future policy interest rate changes. In other words, a rate change may be taken as a signal of the future stance of monetary policy—contractionary or expansionary.

2. Which of the following does a central bank seek to influence directly via the setting of its official interest rate?

A. Import prices
B. Domestic inflation
C. Inflation expectations

Hide Solution

Solution:

C is correct. By setting its official interest rate, a central bank could expect to have a direct influence on inflation expectations—as well as on other market interest rates, asset prices, and the exchange rate (where this is freely floating). If it can influence these factors, it might ultimately hope to influence import prices (via changes in the exchange rate) and also domestically generated inflation (via its impact on domestic or external demand). The problem is that the workings of the transmission mechanism—from the official interest rate to inflation—are complex and can change over time.

3. Monetary policy is *least likely* to include:

A. setting an inflation rate target.
B. changing an official interest rate.
C. enacting a transfer payment program.

Hide Solution

Solution:

C is correct. Transfer payment programs represent fiscal, not monetary policy.

4. Which is the *most* accurate statement regarding central banks and monetary policy?

A. Central bank activities are typically intended to maintain price stability.
B. Monetary policies work through the economy via four independent channels.
C. Commercial and interbank interest rates move inversely to official interest rates.

Hide Solution

Solution:

A is correct. Central bank activities are typically intended to maintain price stability. B is not correct because the transmission channels of monetary policy are not independent.

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